University of Minnesota custom Azure course

Session 5 Participant Workbook

Next-Generation Protection for Oracle and Enterprise Databases

Azure Storage, Data Protection, and Resilience at Scale

Session: Day 3, Session 5, four hours

Use: Checkpoint discussion, design exercise or lab debrief, discussion prompts, knowledge check

Keep: Your worksheets are the input to the Session 6 roadmap capstone

All scenarios use synthetic data modeled on a large public research university. No real University of Minnesota data is used.

Contents

1. Session map 3
1.1 What this session covers 3
1.2 Have ready 3
1.3 Materials you will receive 3
2. Block 1 checkpoint: choosing the Oracle path 4
3. Design exercise: the immutability tier for the enterprise Oracle estate 5
3.1 Steps 5
3.2 Worksheet: backup targets and immutability 5
3.3 Worksheet: separation of duties and geo-protection 6
3.4 Worksheet: restore testing and cost 7
3.5 Worksheet: the three sentences 8
3.6 What you present 9
3.7 How your design will be critiqued 9
4. Discussion prompts 9
5. Knowledge check 10
6. Takeaways and the capstone 12
7. Notes 12

1. Session map

Day 3, Session 5: Next-Generation Protection for Oracle and Enterprise Databases. Four hours including a fifteen-minute break.

Block	Minutes	Activity
Block 1	55	Oracle on Azure deployment options
Block 2	55	Oracle backup to immutable Blob and the live demonstration
Break	15	
Block 3	40	Other database platforms and restore validation
Block 4	75	Design exercise and leadership pitch review

1.1 What this session covers

The university's largest and most business-critical databases run on Oracle and support enterprise resource planning and identity management. The current protection approach is decades old, and the team has been asked to define a modern strategy centered on immutability. This session covers the two main paths for Oracle on Azure (Oracle AI Database@Azure, formerly Oracle Database@Azure, and Oracle on Azure virtual machines), the backup toolbox (RMAN, Azure Backup, Azure Files, Blob through BlobFuse2), making the backup container immutable and aligning RMAN retention with the WORM interval, geo-protection, separation of duties, and coordinated recovery of ERP and identity tiers.

Block 3 covers the other platforms (SQL Server in Azure VMs, Azure SQL, PostgreSQL flexible server with vaulted backup, MySQL) and restore validation as proof rather than hope. The design exercise produces the immutability tier for the enterprise Oracle estate and three sentences a vice president could use.

1.2 Have ready

- Your Oracle estate at a glance: versions, sizes, where RMAN writes today, and the recovery window in force.
- The Session 3 and 4 workbooks: the vault controls and the resilience tier you assigned to the enterprise database.
- The business case template open if you have a laptop; the exercise fills its first page.

1.3 Materials you will receive

- Session 5 deck; demo script `S05_Demo_Script_RMAN_Immutable_Restore`
- `UMN_Enterprise_Application_Dataset.xlsx` (12,302 synthetic journal lines) with the CONTROL_TOTALS sheet
- Oracle on Azure decision guide reference card and the business case template (Reference Card Pack; Executive Briefing and Business Case Template)

- This workbook (participants) and the facilitator guide (instructor)

2. Block 1 checkpoint: choosing the Oracle path

Three questions about the university's Oracle estate. Answer for the ERP and identity databases as you know them.

Question 1

Oracle AI Database@Azure or Oracle on Azure VMs: which path fits the ERP database, and what drives the choice?

Question 2

Why back up a database whose storage is already triple-replicated?

Question 3

Where do licensing and support boundaries belong in the project plan?

3. Design exercise: the immutability tier for the enterprise Oracle estate

In groups, design the target-state protection architecture for the university's Oracle estate: backup targets, immutability policy and retention, separation of duties, restore testing cadence, and estimated monthly cost. The estate: the ERP database (12 TB, Oracle 19c, two instances, nightly change about 3 percent, recovery window required 35 days, RPO 15 minutes, RTO 4 hours), the identity platform database (2 TB, RPO 15 minutes, RTO 2 hours), and four departmental databases (200 GB to 1.5 TB, RPO 24 hours, RTO 24 hours). Assume Oracle on Azure VMs for the ERP and identity databases in the first phase, with Database@Azure evaluated for the second phase.

Each group also drafts three sentences a vice president could use to explain why the business should care, and presents design and pitch for review against Microsoft and Oracle reference guidance.

3.1 Steps

1. Complete Worksheet 1: backup targets and immutability per database class.
2. Complete Worksheet 2: separation of duties (roles, not people) and the geo-protection path.
3. Complete Worksheet 3: restore testing cadence and the estimated monthly cost line, using the cost model workbook's BackupRetention sheet and illustrative prices.
4. Draft the three vice president sentences in Worksheet 4.
5. Present design and pitch (five minutes); the facilitator reviews against the Oracle on Azure decision guide and Microsoft Learn's Oracle backup guidance.

3.2 Worksheet: backup targets and immutability

Decision	ERP database (12 TB)	Identity database (2 TB)	Departmental databases
Backup engine and target (RMAN to Blob via BlobFuse2, Azure Files, Azure Backup)			

Decision	ERP database (12 TB)	Identity database (2 TB)	Departmental databases
Account type, namespace, redundancy for the target			
Immutability: container-level or version-level, interval, locked or unlocked			
RMAN retention policy aligned to the WORM interval			
Where the pieces go after the window (lifecycle, Archive)			

3.3 Worksheet: separation of duties and geo-protection

Item	Your answer
Who runs RMAN; how the mount authenticates (managed identity)	
Who owns the immutable target (container and policy)	

Item	Your answer
Who holds the lock decision and the Resource Guard for vault-based backups	
Geo-protection: GZRS, object replication to an immutable destination, Data Guard, or a combination	
Separate subscription for the backup account: yes or no, and why	
What is audited and who receives alerts	

3.4 Worksheet: restore testing and cost

Item	ERP	Identity	Departmental
Automated restore test: cadence and target environment			
Full recovery rehearsal cadence and what it updates (runbook, measured RTO)			

Item	ERP	Identity	Departmental
Evidence kept (queries, control totals, job logs)			
Estimated monthly cost: capacity in the WORM container, geo copy, vault fees			

3.5 Worksheet: the three sentences

Sentence	Draft
1. What changes (in risk language)	
2. What it costs and what it avoids	
3. What we will be able to prove, and when	

3.6 What you present

- The target-state architecture for the three database classes.
- The separation-of-duties table and the geo-protection path.
- The restore cadence, the cost line, and the three sentences.

3.7 How your design will be critiqued

Lens	Meets	Exceeds
Immutability	RMAN lands in a WORM container; retention aligned to the interval	Lock decision owned by Security; version-level policy where object-lock semantics are needed
Separation of duties	DBA, storage, and Security roles distinct; managed identity for the mount	Backup account in a separate subscription; Resource Guard for vault-based backups
Geo-protection	A second region copy of the backups stated	Replicate while online; immutable destination; Data Guard for the RPO target
Restore validation	Cadence stated per class with an isolated target	Control totals and hashes as evidence; measured RTO against the target
Business case	Three sentences in risk language with a cost	Two or three options with trade-offs

4. Discussion prompts

Work through these in your group or with a neighbor during the time the facilitator calls out. Write enough that you can defend the position in the room.

Prompt 1

RMAN thinks it writes to a directory. What are the consequences of that abstraction for immutability, and what breaks if the mount's identity loses its role?

Block 2.

Prompt 2

Azure SQL and PostgreSQL flexible server have platform-native backups with long-term retention. What does the Oracle estate lose by not being on a managed service, and what does it keep?

Block 3.

Prompt 3

What evidence would you put in front of an auditor to prove restore capability, and how often is it generated?

Block 3.

5. Knowledge check

Six questions, one best answer each. Mark your answer; the facilitator reviews them at the end of the session.

1. What is the current product name for Oracle's database services colocated in Azure datacenters?

- A. Oracle Cloud at Azure
- B. Oracle AI Database@Azure
- C. Azure Database for Oracle
- D. Exadata on Azure

2. Why is NFS 3.0 on Blob not the immutable RMAN target?

- A. It is too slow
- B. Immutability policies are not supported on NFS 3.0-enabled accounts, which also require hierarchical namespace
- C. RMAN cannot write to NFS
- D. It requires premium storage

3. The RMAN recovery window is 30 days. What is the minimum WORM interval on the backup container?

- A. 7 days
- B. 14 days
- C. 30 days
- D. There is no relationship

4. Which Azure service provides long-term retention up to ten years in a Backup vault with immutability for an open-source database?

- A. Azure Database for MySQL
- B. Azure Database for PostgreSQL flexible server
- C. Azure Cache for Redis
- D. Azure Cosmos DB for MongoDB

5. What is the short-term retention range for Azure SQL automatic backups?

- A. 1 to 7 days
- B. 1 to 35 days
- C. 7 to 90 days
- D. 30 to 365 days

6. Who should hold the lock decision for the RMAN container's WORM policy?

- A. The DBA who runs RMAN

- B. The storage administrator who owns the account
- C. Security, as a change-controlled decision
- D. Whoever is on call

6. Takeaways and the capstone

- Choose the Oracle path by operating model, performance, and licensing; bring licensing into the business case early.
- RMAN to immutable Blob through BlobFuse2 on a flat account, with the recovery window aligned to the WORM interval and separation of duties across DBA, storage, and Security, is the immutability tier.
- Replicate backup pieces while online, to an immutable destination in a separate subscription; use Data Guard for the RPO, not Site Recovery, for the database itself.
- Restore validation turns protection into evidence: automate weekly, rehearse quarterly, measure RTO, keep the control totals.

Capstone input from this session

Immutability tier for the Oracle estate.

Next: Session 6, FinOps for research data, data management as code, and the roadmap capstone: every session's cost line added up, the platform as code, and the two-year roadmap.

7. Notes

Your notes